

Homework 6

1. A company sells two types of items A and B . Item A sells for Rs 25 per unit. No quantity discount is given. The sales revenue for item B decreases as the number of its units sold increases. It is given by:

$$\text{Sales revenue} = (30 - 0.30x_2)x_2 = 30x_2 - 0.30x_2^2$$

where x_2 is the number of units sold of item B . The marketing department has only 1200 hours available for distributing these items in the next year. Further, the company estimates the sales time function is non-linear and is given by:

$$\text{Sales time} = x_1 + 0.2x_1^2 + 3x_2 + 0.35x_2^2$$

The company can only procure 6000 units of item A and B for sales in the next year. Given the above information, how many units of item A and B should the company procure in order to maximize its total revenue?

Formulate it as a nonlinear programming problem.

Solution:

The mathematical model of the problem can be stated as follows:

$$\text{Maximize } Z = 25x_1 + 30x_2 - 0.30x_2^2$$

subject to the constraints

$$x_1 + 0.2x_1^2 + 3x_2 + 0.35x_2^2 \leq 1200,$$

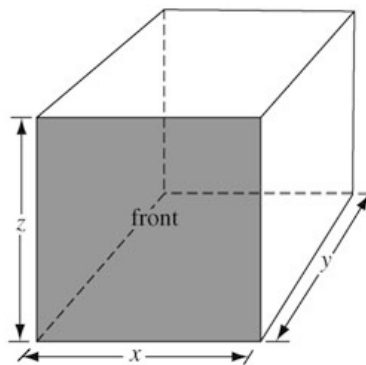
and

$$x_1 + x_2 \leq 6000$$

and

$$x_1, x_2 \geq 0$$

2. A cardboard box for packing quantities of small foam balls is to be manufactured as shown in the following figure. The top, bottom, and front faces must be of double weight (i.e., two pieces of cardboard). A problem posed is to find the dimensions of such a box that maximize the volume for a given amount of cardboard, equal to 72 sq. ft.



Formulate it as a nonlinear programming problem.

Solution:

$$\begin{array}{ll}\max & Z = xyz \\ \text{s.t.} & 4xy + 3xz + 2yx \leq 72 \\ & x, y, z \geq 0\end{array}$$

3.

Solve the following problem by using the method of Lagrangian multipliers.

$$\text{Minimize } Z = x_1^2 + x_2^2 + x_3^2$$

subject to the constraints

$$\begin{aligned}x_1 + x_2 + 3x_3 &= 2 \\ 5x_1 + 2x_2 + x_3 &= 5\end{aligned}$$

and

$$x_1, x_2 \geq 0$$

Solution: The Lagrangian function is

$$L(\mathbf{x}, \lambda) = x_1^2 + x_2^2 + x_3^2 - \lambda_1 (x_1 + x_2 + 3x_3 - 2) - \lambda_2 (5x_1 + 2x_2 + x_3 - 5)$$

The necessary conditions for the minimum of Z give us the following:

$$\begin{aligned}\frac{\partial L}{\partial x_1} &= 2x_1 - \lambda_1 - 5\lambda_2 = 0; & \frac{\partial L}{\partial x_2} &= 2x_2 - \lambda_1 - 2\lambda_2 = 0 \\ \frac{\partial L}{\partial x_3} &= 2x_3 - 3\lambda_1 - \lambda_2 = 0 \\ \lambda_i g_i &= 0, & \lambda_1 (x_1 + x_2 + 3x_3 - 2), \lambda_2 (5x_1 + 2x_2 + x_3 - 5) &= 0 \\ x_1 + x_2 + 3x_3 - 2 &= 0, & 5x_1 + 2x_2 + x_3 - 5 &= 0, x_1, x_2 \geq 0 \\ \lambda_i &\text{is free}\end{aligned}$$

The solution of these simultaneous equations gives:

$$\mathbf{x} = (x_1, x_2, x_3) = (37/46, 16/46, 13/46); \lambda = (\lambda_1, \lambda_2) = (2/23, 7/23) \text{ and } Z = 897/1058$$

4.(additional)

Determine x_1 and x_2 so as to

$$\text{Maximize } Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$$

subject to the constraints

$$\begin{aligned}x_2 &\leq 8, \\ x_1 + x_2 &\leq 10\end{aligned}$$

and

$$x_1, x_2 \geq 0$$

Solution:

$$\begin{aligned}\text{Here } f(x_1, x_2) &= 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2 \\ g_1(x_1, x_2) &= x_2 - 8 \leq 0 \\ g_2(x_1, x_2) &= x_1 + x_2 - 10 \leq 0\end{aligned}$$

The Lagrangian function can be formulated as:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(x) - \lambda_1 [g_1(x)] - \lambda_2 [g_2(x)]$$

The Kuhn-Tucker necessary conditions can be stated as:

(i)

$$\begin{aligned}\frac{\partial f}{\partial x_j} - \sum_{i=1}^2 \lambda_i \frac{\partial g_i}{\partial x_j}, \quad j = 1, 2 \\ 12 + 2x_2 - 4x_1 - \lambda_2 = 0 \\ 21 + 2x_1 - 4x_2 - \lambda_1 - \lambda_2 = 0\end{aligned}$$

(ii)

$$\begin{aligned}\lambda_i g_i(x) = 0, \quad i = 1, 2 \\ \lambda_1 (x_2 - 8) = 0 \\ \lambda_2 (x_1 + x_2 - 10) = 0\end{aligned}$$

(iii)

$$\begin{aligned}g_i(x) \leq 0 \\ x_2 - 8 \leq 0 \\ x_1 + x_2 - 10 \leq 0 \\ x_1, x_2 \geq 0\end{aligned}$$

(iv)

$$\lambda_i \geq 0, \quad i = 1, 2$$

There may arise four cases: Case 1: If $\lambda_1 = 0, \lambda_2 = 0$, then from Condition (i), we have:

$$12 + 2x_2 - 4x_1 = 0 \text{ and } 21 + 2x_1 - 4x_2 = 0$$

Solving these equations, we get $x_1 = 15/2, x_2 = 9$. However, this solution violates condition (iii) and therefore it should be discarded.

Case 2: $\lambda_1 \neq 0, \lambda_2 \neq 0$, then from condition (ii) we have:

$$\begin{aligned}x_2 - 8 = 0 \quad \text{or} \quad x_2 = 8 \\ x_1 + x_2 - 10 = 0 \quad \text{or} \quad x_1 = 2\end{aligned}$$

Substituting these values in condition (i), we get $\lambda_1 = -27$ and $\lambda_2 = 20$. However, this solution violates the condition (iv) and therefore may be discarded.

Case 3: $\lambda_1 \neq 0, \lambda_2 = 0$, then from conditions (ii) and (i) we have:

$$\begin{aligned}x_1 + x_2 = 10 \\ 2x_2 - 4x_1 = -12 \\ 2x_1 - 4x_2 = -12 + \lambda_1\end{aligned}$$

Solving these equations, we get $x_1 = 2, x_2 = 8$ and $\lambda_1 = -16$. However, this solution violates the condition (iv) and therefore may be discarded.

Case 4: $\lambda_1 = 0, \lambda_2 \neq 0$, then from conditions (i) and (ii) we have:

$$\begin{aligned}2x_2 - 4x_1 = -12 + \lambda_2 \\ 2x_1 - 4x_2 = -21 + \lambda_2 \\ x_1 + x_2 = 10\end{aligned}$$

Solving these equations, we get $x_1 = 17/4, x_2 = 23/4$ and $\lambda_2 = 13/2$. This solution does not violate any of the Kuhn-Tucker conditions and therefore must be accepted.

Hence, the optimum solution of the given problem is: $x_1 = 17/4, x_2 = 23/4, \lambda_1 = 0$ and $\lambda_2 = 13/2$ and $\text{Max } Z = 947/8$.